

GRP - 02

STATISTICAL MACHINE TRANSLATION

English to Hindi

Team Members



Yash Akotkar

Roll no.: 371006



Rajdeep Chaurasia

Roll no.: 371015



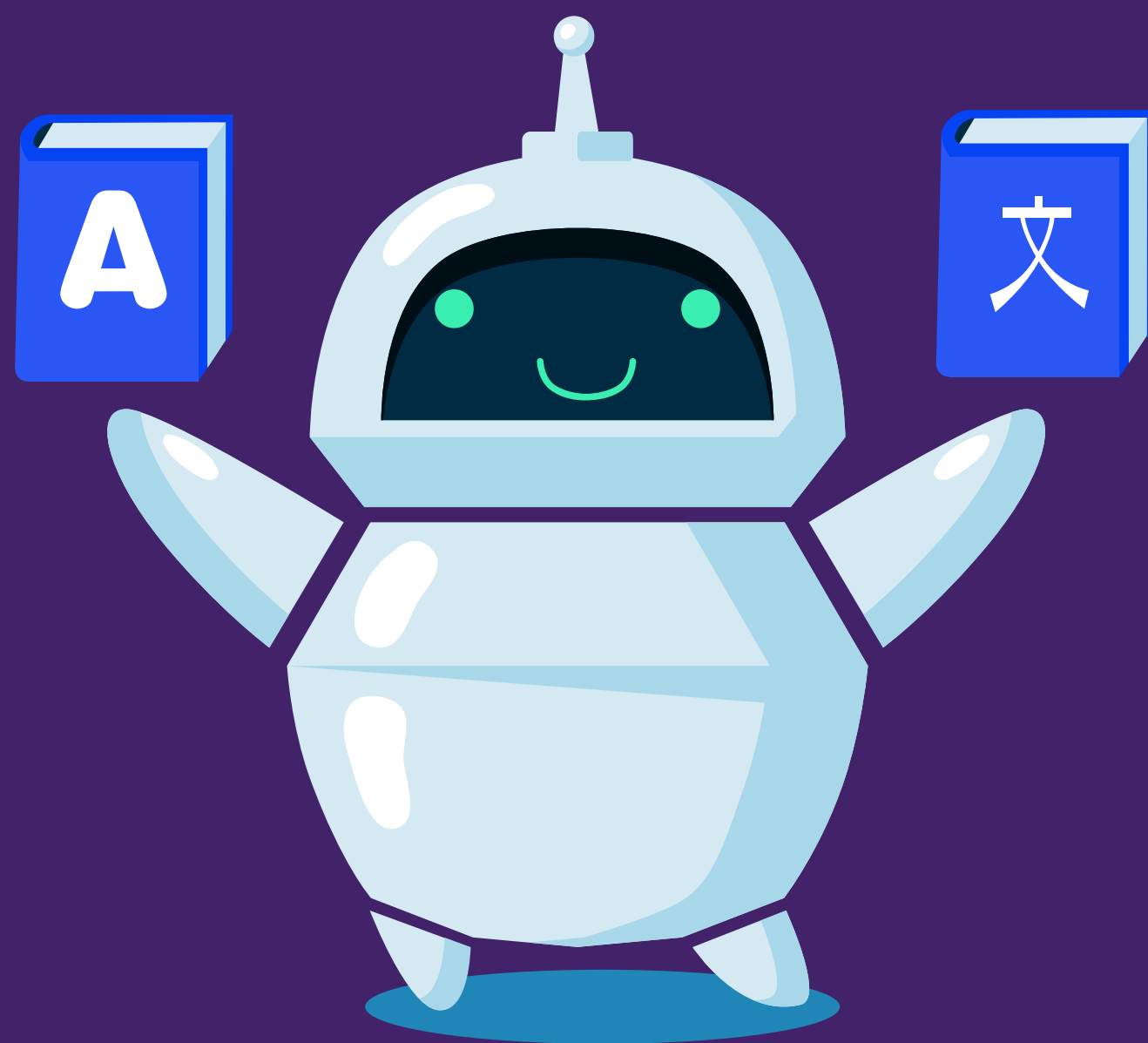
Netal Daga

Roll no.: 371016



Samarth Ghule

Roll no.: 371019



Introduction

Statistical machine translation, known as SMT or StatMT, is an approach to machine translation which yields the most probable output (translation) of each element that makes up a sentence. It is based on the use of statistical models that analyze and search for relationships between two texts with the same content: one in the source language and the other in the target language.

A text is translated based on the probability that a string of words in the target language is the translation of the string of words in the source language.



Approach

It just requires data in the form of example translations from which a translation model may be learned. We determine the string of words(f^*) that maximizes:

$$f^* = \operatorname{argmax}_f P(f | e) = \operatorname{argmax}_f P(e | f)P(f)$$

- The target language model for Hindi is " $P(f)$ ", which indicates how likely a particular sentence is in Hindi.
- The translation model, $P(e|f)$, indicates how likely an English sentence is to be translated into a
- particular Hindi sentence. Similarly, $P(f|e)$ is an English-to-Hindi translation model.

e_1	e_2	e_3	e_4	e_5
There is a	smelly	wump us	sleeping	in 22
I1 ya un	Wumpus	malo dirant	quidort	a 2 2
$D_1 = 0$	$D_3 = -2$	$D_2 = +1$	$D_4 = +1$	$D_5 = 0$

Given an English sentence (e), finding a Hindi translation $f()$ is a three-step process:

1. Divide the English sentence into $e_1, \dots, e_n$.

2. Choose a French phrase f_i for each phrase e_i . For the phrasal probability that f_i is a translation of e_i , we use the notation $P(f_i | e_i)$.

3. Pick a combination of the terms $f_1, \dots, f_n$. This permutation will be specified in a style that appears difficult but is supposed to have a simple probability distribution:

- We pick a distortion d_i for each f_i , which is the number of words that phrase f_i has moved with regard to f_{i-1} ; positive for travelling to the right, negative for moving to the left, and zero if f_i follows f_{i-1} directly.

Dataset

	english_sentence	hindi_sentence
0	politicians do not have permission to do what needs to be done.	"राजनीतिज्ञों के पास जो कार्य करना चाहिए, वह करने कि अनुमति नहीं है."
1	"I'd like to tell you about one such child,"	"मई आपको ऐसे ही एक बच्चे के बारे में बताना चाहूंगी,"
2	This percentage is even greater than the percentage in India.	यह प्रतिशत भारत में हिन्दुओं प्रतिशत से अधिक है।
3	what we really mean is that they're bad at not paying attention.	हम ये नहीं कहना चाहते कि वो ध्यान नहीं दे पाते
4	.The ending portion of these Vedas is called Upanishad.	इन्हीं वेदों का अंतिम भाग उपनिषद कहलाता है।
5	"The then Governor of Kashmir resisted transfer , but was finally reduced to subjection with the aid of British ."	
6	In this lies the circumstances of people before you.	इसमें तुमसे पूर्व गुज़रे हुए लोगों के हालात हैं।
7	"And who are we to say, even, that they are wrong"	और हम होते कौन हैं यह कहने भी वाले कि वे गलत हैं
8	""Global Warming"" refer to warming caused in recent decades and probability of its continual presence and its indirect	
9	You may want your child to go to a school that is not run by the LEA - a non-maintained special school or an independen	
10	Please ensure that you use the appropriate form .	कृपया यह सुनिश्चित कर लें कि आप सही फॉर्म का प्रयोग कर रहे हैं .
11	Category: Religious Text	श्रेणी:धर्मग्रन्थ
12	This period summarily is pepped up with devotion.	यह काल समग्रतः भक्ति भावना से ओतप्रोत काल है।
13	So there is some sort of justice	तो वहाँ न्याय है
14	The first two were found unreliable and the prosecution case rested mainly on the evidence of the remaining five approv	
15	They had justified their educational policy of concentrating on the education of a small number of upper and middle-cla	
16	"And now at present the naturecure, Ayurvedic and modern treatments are taking place through the government in Nepal."	
17	Parliament time frame is 5 years and this will be dissolved before that.	लोकसभा की कार्यविधि 5 वर्ष है परंतु इसे समय से पूर्व भंग किया जा स
18	"ii Register Courts , empowered to try causes for amounts not exceeding Rs 200 , when authorised by the judges ."	
19	"Extreme weather due to increased mortality; displacements and economic loss will be compounded through growing populat	
20	Of these Lahadi is a popular one .	' लाडड़ी लोकप्रिय स्त्री नृत्य है .
21	Even a concentration of 0.001 ppm of hydrogen sulphide in the water can emit the smell of rotten egg .	यहां तक कि पानी में हाइड्रोज
22	"Islam is the world's second-largest religion, after Christianity."	इस्लाम धर्म (الإسلام) ईसाई धर्म के बाद अनुयाइयों के आधार पर दुनिया का दूसरा सब र
23	This changed slowly	धीरे धीरे ये सब बदला

Total Rows: 177607,
Columns: 2



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Implementation

Output

```
k+=1
(input_seq, actual_output), _ = next(train_gen)
decoded_sentence = decode_sequence(input_seq)
print('Input English sentence:', X_train[k:k+1].values[0])
print('Actual Hindi Translation:', y_train[k:k+1].values[0][6:-4])
print('Predicted Hindi Translation:', decoded_sentence[:-4])
```

[47]

... Input English sentence: hindu philosophy of unity
Actual Hindi Translation: हिन्दुत्व एकत्व का दर्शन है
Predicted Hindi Translation: भारत में एक प्रकार

```
k+=1
(input_seq, actual_output), _ = next(train_gen)
decoded_sentence = decode_sequence(input_seq)
print('Input English sentence:', X_train[k:k+1].values[0])
print('Actual Hindi Translation:', y_train[k:k+1].values[0][6:-4])
print('Predicted Hindi Translation:', decoded_sentence[:-4])
```

[33]

... Input English sentence: and god“ he said to the poor man “is answering your plea
Actual Hindi Translation: “और भगवान्” उन्होंने गरीब व्यक्ति से कहा “तुम्हारी प्रार्थना सुन रहे हैं
Predicted Hindi Translation: और यह एक बार एक तरह से एक तरह से नहीं है



STUDIO SHODWE



THANK YOU

